

PCBC H Vectors & Dot Product: Memorize Sheet

everything below is from memory on quiz day

Component form

$PQ = \langle x_2 - x_1, y_2 - y_1 \rangle$ (terminal - initial)

$\langle a, b \rangle = ai + bj$, $\mathbf{i} = \langle 1, 0 \rangle$, $\mathbf{j} = \langle 0, 1 \rangle$

$|\langle a, b \rangle| = \sqrt{a^2 + b^2}$

unit vector: magnitude 1

Dot product

$\mathbf{u} \cdot \mathbf{v} = ac + bd$ (a number, not a vector)

$\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$

$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

$(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (k\mathbf{v})$

Magnitude and direction

$a = |\mathbf{v}|\cos \theta$ (horizontal)

$b = |\mathbf{v}|\sin \theta$ (vertical)

$\mathbf{v} = \langle |\mathbf{v}|\cos \theta, |\mathbf{v}|\sin \theta \rangle$

$\tan \theta = b/a$, $a \neq 0$ (reference angle)

report θ in $[0, 2\pi)$

Angle between

$\cos \theta = (\mathbf{u} \cdot \mathbf{v}) / (|\mathbf{u}||\mathbf{v}|)$, $0^\circ \leq \theta \leq 180^\circ$

orthogonal $\iff \mathbf{u} \cdot \mathbf{v} = 0$

dot > 0 : acute

dot $= 0$: right

dot < 0 : obtuse

Quadrant fix

$\tan^{-1}$ only answers in QI or QIV. signs of $\langle a, b \rangle$ give the quadrant:

QI (+, +): $\theta = \text{ref}$

QII (-, +): $\theta = 180^\circ - \text{ref}$

QIII (-, -): $\theta = 180^\circ + \text{ref}$

QIV (+, -): $\theta = 360^\circ - \text{ref}$

Two forces at a point

angle θ between them $\Rightarrow$ use the supplement $180^\circ - \theta$ in the triangle

$x = \sqrt{a^2 + b^2 - 2ab \cos C}$

one radical, round at the very end

Operations

$\langle a, b \rangle \pm \langle c, d \rangle = \langle a \pm c, b \pm d \rangle$

$k\langle a, b \rangle = \langle ka, kb \rangle$

$|k\mathbf{v}| = |k||\mathbf{v}|$ (the sign never shortens it)

resultant = the sum: tip to tail, or the diagonal of the parallelogram

addition commutes, subtraction does not

Exact values

θ	$\sin \theta$	$\cos \theta$
0	0	1
$\pi/6 = 30^\circ$	1/2	$\sqrt{3}/2$
$\pi/4 = 45^\circ$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3 = 60^\circ$	$\sqrt{3}/2$	1/2
$\pi/2 = 90^\circ$	1	0

signs by quadrant (ASTC): QI all +; QII sin +; QIII tan +; QIV cos +

Non-Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find $3\mathbf{u} + \mathbf{v} - 5\mathbf{v}$ and $|2\mathbf{u} - \mathbf{v}|$ for the given vectors $\mathbf{u}$ and $\mathbf{v}$. Write $3\mathbf{u} + \mathbf{v} - 5\mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 points)

1. $\mathbf{u} = -3\mathbf{i} + 2\mathbf{j}$, $\mathbf{v} = 4\mathbf{i} - \mathbf{j}$

Express the vector with initial point P and terminal point Q in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. Then find $|-2\mathbf{PQ}|$. (2 points)

2. P(6, -2), Q(-2, 4)

Find the horizontal and vertical components of each vector. Then write $\mathbf{u} + \mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{u} + \mathbf{v}|$. (3 points)

3. $|\mathbf{u}| = 8, \theta = \pi; |\mathbf{v}| = 4, \theta = \pi/3$

Find the magnitude and direction of the vector. Write your answer in radians. (3 points)

4. $\mathbf{u} = -(\sqrt{3}/2)\mathbf{i} + (1/2)\mathbf{j}$

Find $(2\mathbf{u}) \cdot \mathbf{v}$ and $\mathbf{u} \cdot (\mathbf{u} + \mathbf{v})$ for the given vectors. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and state whether the angle between them is acute, right, or obtuse. (3 points)

5. $\mathbf{u} = 4\mathbf{i}, \mathbf{v} = -3\mathbf{i} + 5\mathbf{j}$

Calculator

Find the angle between the given vectors. Round to the nearest tenth of a degree. (3 points)

6. $\mathbf{u} = 3\mathbf{i} + 4\sqrt{2}\mathbf{j}$, $\mathbf{v} = \sqrt{2}\mathbf{i} - \mathbf{j}$

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force. Round to the nearest tenth. (3 points)

7. forces of 240 N and 175 N, forming an angle of 63°

ANSWER KEY

Find $3\mathbf{u} + \mathbf{v} - 5\mathbf{v}$ and $|2\mathbf{u} - \mathbf{v}|$ for the given vectors $\mathbf{u}$ and $\mathbf{v}$. Write $3\mathbf{u} + \mathbf{v} - 5\mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 points)

1. $\mathbf{u} = -3\mathbf{i} + 2\mathbf{j}$, $\mathbf{v} = 4\mathbf{i} - \mathbf{j}$

collect first: $\mathbf{v} - 5\mathbf{v} = -4\mathbf{v}$, so the expression is $3\mathbf{u} - 4\mathbf{v}$

$\mathbf{u} = \langle -3, 2 \rangle$, $\mathbf{v} = \langle 4, -1 \rangle$; $3\mathbf{u} = \langle -9, 6 \rangle$, $4\mathbf{v} = \langle 16, -4 \rangle$

$3\mathbf{u} + \mathbf{v} - 5\mathbf{v} = \langle -9 - 16, 6 - (-4) \rangle = \langle -25, 10 \rangle = -25\mathbf{i} + 10\mathbf{j}$

$2\mathbf{u} - \mathbf{v} = \langle -6 - 4, 4 - (-1) \rangle = \langle -10, 5 \rangle$

$|2\mathbf{u} - \mathbf{v}| = \sqrt{(100 + 25)} = \sqrt{125} = 5\sqrt{5}$

Express the vector with initial point P and terminal point Q in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. Then find $|-2\mathbf{PQ}|$. (2 points)

2. P(6, -2), Q(-2, 4)

$\mathbf{PQ} = \langle -2 - 6, 4 - (-2) \rangle = \langle -8, 6 \rangle = -8\mathbf{i} + 6\mathbf{j}$

$|\mathbf{PQ}| = \sqrt{(64 + 36)} = 10$

scaling multiplies the length by $|k|$: $|-2\mathbf{PQ}| = 2(10) = 20$ (the sign does not shorten it)

Find the horizontal and vertical components of each vector. Then write $\mathbf{u} + \mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{u} + \mathbf{v}|$. (3 points)

3. $|\mathbf{u}| = 8$, $\theta = \pi$; $|\mathbf{v}| = 4$, $\theta = \pi/3$

$\mathbf{u} = \langle 8\cos \pi, 8\sin \pi \rangle = \langle 8(-1), 8(0) \rangle = \langle -8, 0 \rangle$ (horizontal -8, vertical 0)

$\mathbf{v} = \langle 4\cos(\pi/3), 4\sin(\pi/3) \rangle = \langle 2, 2\sqrt{3} \rangle$

$\mathbf{u} + \mathbf{v} = \langle -8 + 2, 0 + 2\sqrt{3} \rangle = \langle -6, 2\sqrt{3} \rangle = -6\mathbf{i} + 2\sqrt{3}\mathbf{j}$

$|\mathbf{u} + \mathbf{v}| = \sqrt{(36 + 12)} = \sqrt{48} = 4\sqrt{3}$

Find the magnitude and direction of the vector. Write your answer in radians. (3 points)

4. $\mathbf{u} = -(\sqrt{3}/2)\mathbf{i} + (1/2)\mathbf{j}$

$|\mathbf{u}| = \sqrt{(3/4 + 1/4)} = \sqrt{1} = 1$ (a unit vector, the fractions are not a mistake)

$\tan \theta = (1/2)/(-\sqrt{3}/2) = -1/\sqrt{3} \Rightarrow$ reference angle $= \pi/6$

$a < 0$, $b > 0 \Rightarrow$ QII, so $\theta = \pi - \pi/6 = 5\pi/6$

Find $(2\mathbf{u}) \cdot \mathbf{v}$ and $\mathbf{u} \cdot (\mathbf{u} + \mathbf{v})$ for the given vectors. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and state whether the angle between them is acute, right, or obtuse. (3 points)

5. $\mathbf{u} = 4\mathbf{i}, \mathbf{v} = -3\mathbf{i} + 5\mathbf{j}$

the missing component is a zero: $\mathbf{u} = \langle 4, 0 \rangle, \mathbf{v} = \langle -3, 5 \rangle$

$$\mathbf{u} \cdot \mathbf{v} = (4)(-3) + (0)(5) = -12$$

$$(2\mathbf{u}) \cdot \mathbf{v} = 2(\mathbf{u} \cdot \mathbf{v}) = \boxed{-24}$$

$$\mathbf{u} + \mathbf{v} = \langle 1, 5 \rangle \Rightarrow \mathbf{u} \cdot (\mathbf{u} + \mathbf{v}) = (4)(1) + (0)(5) = \boxed{4}$$

$\mathbf{u} \cdot \mathbf{v} = -12 \neq 0$, so not orthogonal; negative $\Rightarrow$ obtuse

Find the angle between the given vectors. Round to the nearest tenth of a degree. (3 points)

6. $\mathbf{u} = 3\mathbf{i} + 4\sqrt{2}\mathbf{j}, \mathbf{v} = \sqrt{2}\mathbf{i} - \mathbf{j}$

$$\mathbf{u} \cdot \mathbf{v} = (3)(\sqrt{2}) + (4\sqrt{2})(-1) = 3\sqrt{2} - 4\sqrt{2} = -\sqrt{2}$$

$$|\mathbf{u}| = \sqrt{9 + 32} = \sqrt{41}, |\mathbf{v}| = \sqrt{2 + 1} = \sqrt{3}$$

$$\cos \theta = (-\sqrt{2})/(\sqrt{41}\sqrt{3}) \approx -0.1275$$

$$\theta = \cos^{-1}(-0.1275) \approx \boxed{97.3^\circ} \text{ (negative dot product, so obtuse } \checkmark)$$

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force. Round to the nearest tenth. (3 points)

7. forces of 240 N and 175 N, forming an angle of 63°

the resultant is the diagonal; the angle across from it is $180^\circ - 63^\circ = 117^\circ$

$$x = \sqrt{240^2 + 175^2 - 2(240)(175)\cos 117^\circ}$$

$$x \approx \boxed{355.5 \text{ N}}$$

Non-Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find $4\mathbf{v} + \mathbf{u} - 3\mathbf{u}$ and $|\mathbf{u} + 2\mathbf{v}|$ for the given vectors $\mathbf{u}$ and $\mathbf{v}$. Write $4\mathbf{v} + \mathbf{u} - 3\mathbf{u}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 points)

1. $\mathbf{u} = 5\mathbf{i} - 2\mathbf{j}$, $\mathbf{v} = -\mathbf{i} + 3\mathbf{j}$

Express the vector with initial point P and terminal point Q in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. Then find $|\overrightarrow{PQ}|$. (2 points)

2. P(-5, 3), Q(7, -2)

Find the horizontal and vertical components of each vector. Then write $\mathbf{u} + \mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{u} + \mathbf{v}|$. (3 points)

3. $|\mathbf{u}| = 6, \theta = 3\pi/2; |\mathbf{v}| = 10, \theta = 5\pi/6$

Find the magnitude and direction of the vector. Write your answer in radians. (3 points)

4. $\mathbf{u} = (1/2)\mathbf{i} - (\sqrt{3}/2)\mathbf{j}$

Find $(-3\mathbf{u}) \cdot \mathbf{v}$ and $\mathbf{v} \cdot (\mathbf{u} + \mathbf{v})$ for the given vectors. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and state whether the angle between them is acute, right, or obtuse. (3 points)

5. $\mathbf{u} = -6\mathbf{j}, \mathbf{v} = 2\mathbf{i} - 5\mathbf{j}$

Calculator

Find the angle between the given vectors. Round to the nearest tenth of a degree. (3 points)

6. $\mathbf{u} = -2\mathbf{i} + 5\sqrt{2}\mathbf{j}$, $\mathbf{v} = 3\sqrt{2}\mathbf{i} + \mathbf{j}$

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force. Round to the nearest tenth. (3 points)

7. forces of 310 lb and 145 lb, forming an angle of 48°

ANSWER KEY

Find $4\mathbf{v} + \mathbf{u} - 3\mathbf{u}$ and $|\mathbf{u} + 2\mathbf{v}|$ for the given vectors $\mathbf{u}$ and $\mathbf{v}$. Write $4\mathbf{v} + \mathbf{u} - 3\mathbf{u}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 points)

1. $\mathbf{u} = 5\mathbf{i} - 2\mathbf{j}$, $\mathbf{v} = -\mathbf{i} + 3\mathbf{j}$

collect first: $\mathbf{u} - 3\mathbf{u} = -2\mathbf{u}$, so the expression is $4\mathbf{v} - 2\mathbf{u}$

$\mathbf{u} = \langle 5, -2 \rangle$, $\mathbf{v} = \langle -1, 3 \rangle$; $4\mathbf{v} = \langle -4, 12 \rangle$, $2\mathbf{u} = \langle 10, -4 \rangle$

$4\mathbf{v} + \mathbf{u} - 3\mathbf{u} = \langle -4 - 10, 12 - (-4) \rangle = \langle -14, 16 \rangle = -14\mathbf{i} + 16\mathbf{j}$

$\mathbf{u} + 2\mathbf{v} = \langle 5 + (-2), -2 + 6 \rangle = \langle 3, 4 \rangle$

$|\mathbf{u} + 2\mathbf{v}| = \sqrt{9 + 16} = 5$

Express the vector with initial point P and terminal point Q in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. Then find $|3\mathbf{PQ}|$. (2 points)

2. P(-5, 3), Q(7, -2)

$\mathbf{PQ} = \langle 7 - (-5), -2 - 3 \rangle = \langle 12, -5 \rangle = 12\mathbf{i} - 5\mathbf{j}$

$|\mathbf{PQ}| = \sqrt{144 + 25} = \sqrt{169} = 13$

$|3\mathbf{PQ}| = 3(13) = 39$

Find the horizontal and vertical components of each vector. Then write $\mathbf{u} + \mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{u} + \mathbf{v}|$. (3 points)

3. $|\mathbf{u}| = 6$, $\theta = 3\pi/2$; $|\mathbf{v}| = 10$, $\theta = 5\pi/6$

$\mathbf{u} = \langle 6\cos(3\pi/2), 6\sin(3\pi/2) \rangle = \langle 6(0), 6(-1) \rangle = \langle 0, -6 \rangle$ (horizontal 0)

$\mathbf{v} = \langle 10\cos(5\pi/6), 10\sin(5\pi/6) \rangle = \langle -5\sqrt{3}, 5 \rangle$ (QII)

$\mathbf{u} + \mathbf{v} = \langle 0 + (-5\sqrt{3}), -6 + 5 \rangle = \langle -5\sqrt{3}, -1 \rangle = -5\sqrt{3}\mathbf{i} - \mathbf{j}$

$|\mathbf{u} + \mathbf{v}| = \sqrt{75 + 1} = \sqrt{76} = 2\sqrt{19}$

Find the magnitude and direction of the vector. Write your answer in radians. (3 points)

4. $\mathbf{u} = (1/2)\mathbf{i} - (\sqrt{3}/2)\mathbf{j}$

$|\mathbf{u}| = \sqrt{(1/4) + (3/4)} = 1$

$\tan \theta = (-\sqrt{3}/2)/(1/2) = -\sqrt{3} \Rightarrow \text{reference angle} = \pi/3$

$a > 0$, $b < 0 \Rightarrow \text{QIV}$, so $\theta = 2\pi - \pi/3 = 5\pi/3$

Find $(-3\mathbf{u}) \cdot \mathbf{v}$ and $\mathbf{v} \cdot (\mathbf{u} + \mathbf{v})$ for the given vectors. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and state whether the angle between them is acute, right, or obtuse. (3 points)

5. $\mathbf{u} = -6\mathbf{j}$, $\mathbf{v} = 2\mathbf{i} - 5\mathbf{j}$

$$\mathbf{u} = \langle 0, -6 \rangle, \mathbf{v} = \langle 2, -5 \rangle$$

$$\mathbf{u} \cdot \mathbf{v} = (0)(2) + (-6)(-5) = 30$$

$$(-3\mathbf{u}) \cdot \mathbf{v} = -3(30) = \boxed{-90}$$

$$\mathbf{u} + \mathbf{v} = \langle 2, -11 \rangle \Rightarrow \mathbf{v} \cdot (\mathbf{u} + \mathbf{v}) = (2)(2) + (-5)(-11) = \boxed{59}$$

$$\mathbf{u} \cdot \mathbf{v} = 30 \neq 0, \text{ so not orthogonal; positive} \Rightarrow \text{acute}$$

Find the angle between the given vectors. Round to the nearest tenth of a degree. (3 points)

6. $\mathbf{u} = -2\mathbf{i} + 5\sqrt{2}\mathbf{j}$, $\mathbf{v} = 3\sqrt{2}\mathbf{i} + \mathbf{j}$

$$\mathbf{u} \cdot \mathbf{v} = (-2)(3\sqrt{2}) + (5\sqrt{2})(1) = -6\sqrt{2} + 5\sqrt{2} = -\sqrt{2}$$

$$|\mathbf{u}| = \sqrt{4 + 50} = \sqrt{54}, |\mathbf{v}| = \sqrt{18 + 1} = \sqrt{19}$$

$$\cos \theta = (-\sqrt{2})/(\sqrt{54}\sqrt{19}) \approx -0.0442$$

$$\theta = \cos^{-1}(-0.0442) \approx \boxed{92.5^\circ} \text{ (just past a right angle, which the small negative dot product predicts)}$$

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force. Round to the nearest tenth. (3 points)

7. forces of 310 lb and 145 lb, forming an angle of 48°

$$\text{angle across from the diagonal: } 180^\circ - 48^\circ = 132^\circ$$

$$x = \sqrt{310^2 + 145^2 - 2(310)(145)\cos 132^\circ}$$

$$x \approx \boxed{421.0 \text{ lb}}$$

Non-Calculator

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Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find $2\mathbf{u} + 3\mathbf{v} - 5\mathbf{v}$ and $|\mathbf{u} + \mathbf{v}|$ for the given vectors $\mathbf{u}$ and $\mathbf{v}$. Write $2\mathbf{u} + 3\mathbf{v} - 5\mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 points)

1. $\mathbf{u} = 4\mathbf{i} - 3\mathbf{j}$, $\mathbf{v} = -2\mathbf{i} + 6\mathbf{j}$

Express the vector with initial point P and terminal point Q in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. Then find $|-3\mathbf{PQ}|$. (2 points)

2. P(2, -6), Q(-4, 2)

Find the horizontal and vertical components of each vector. Then write $\mathbf{u} + \mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{u} + \mathbf{v}|$. (3 points)

3. $|\mathbf{u}| = 10, \theta = \pi/2; |\mathbf{v}| = 8, \theta = 7\pi/6$

Find the magnitude and direction of the vector. Write your answer in radians. (3 points)

4. $\mathbf{u} = -(1/2)\mathbf{i} - (\sqrt{3}/2)\mathbf{j}$

Find $(3\mathbf{u}) \cdot \mathbf{v}$ and $\mathbf{u} \cdot (\mathbf{u} + \mathbf{v})$ for the given vectors. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and state whether the angle between them is acute, right, or obtuse. (3 points)

5. $\mathbf{u} = -5\mathbf{i}, \mathbf{v} = 4\mathbf{i} - 3\mathbf{j}$

Calculator

Find the angle between the given vectors. Round to the nearest tenth of a degree. (3 points)

6. $\mathbf{u} = 4\mathbf{i} - 3\sqrt{2}\mathbf{j}$, $\mathbf{v} = \sqrt{2}\mathbf{i} + 5\mathbf{j}$

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force. Round to the nearest tenth. (3 points)

7. forces of 88 N and 130 N, forming an angle of 71°

ANSWER KEY

Find $2\mathbf{u} + 3\mathbf{v} - 5\mathbf{v}$ and $|\mathbf{u} + \mathbf{v}|$ for the given vectors $\mathbf{u}$ and $\mathbf{v}$. Write $2\mathbf{u} + 3\mathbf{v} - 5\mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 points)

1. $\mathbf{u} = 4\mathbf{i} - 3\mathbf{j}$, $\mathbf{v} = -2\mathbf{i} + 6\mathbf{j}$

collect first: $3\mathbf{v} - 5\mathbf{v} = -2\mathbf{v}$, so the expression is $2\mathbf{u} - 2\mathbf{v}$

$\mathbf{u} = \langle 4, -3 \rangle$, $\mathbf{v} = \langle -2, 6 \rangle$; $2\mathbf{u} = \langle 8, -6 \rangle$, $2\mathbf{v} = \langle -4, 12 \rangle$

$2\mathbf{u} + 3\mathbf{v} - 5\mathbf{v} = \langle 8 - (-4), -6 - 12 \rangle = \langle 12, -18 \rangle = 12\mathbf{i} - 18\mathbf{j}$

$\mathbf{u} + \mathbf{v} = \langle 2, 3 \rangle \Rightarrow |\mathbf{u} + \mathbf{v}| = \sqrt{4 + 9} = \sqrt{13}$

Express the vector with initial point P and terminal point Q in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$. Then find $|-3\mathbf{PQ}|$. (2 points)

2. P(2, -6), Q(-4, 2)

$\mathbf{PQ} = \langle -4 - 2, 2 - (-6) \rangle = \langle -6, 8 \rangle = -6\mathbf{i} + 8\mathbf{j}$

$|\mathbf{PQ}| = \sqrt{36 + 64} = 10$

$|-3\mathbf{PQ}| = 3(10) = 30$ (length uses $|k|$, so the negative does not matter)

Find the horizontal and vertical components of each vector. Then write $\mathbf{u} + \mathbf{v}$ in component form AND in terms of $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{u} + \mathbf{v}|$. (3 points)

3. $|\mathbf{u}| = 10$, $\theta = \pi/2$; $|\mathbf{v}| = 8$, $\theta = 7\pi/6$

$\mathbf{u} = \langle 10\cos(\pi/2), 10\sin(\pi/2) \rangle = \langle 0, 10 \rangle$ (horizontal 0, so write $0\mathbf{i} + 10\mathbf{j}$)

$\mathbf{v} = \langle 8\cos(7\pi/6), 8\sin(7\pi/6) \rangle = \langle -4\sqrt{3}, -4 \rangle$ (QIII)

$\mathbf{u} + \mathbf{v} = \langle 0 + (-4\sqrt{3}), 10 + (-4) \rangle = \langle -4\sqrt{3}, 6 \rangle = -4\sqrt{3}\mathbf{i} + 6\mathbf{j}$

$|\mathbf{u} + \mathbf{v}| = \sqrt{48 + 36} = \sqrt{84} = 2\sqrt{21}$

Find the magnitude and direction of the vector. Write your answer in radians. (3 points)

4. $\mathbf{u} = -(1/2)\mathbf{i} - (\sqrt{3}/2)\mathbf{j}$

$|\mathbf{u}| = \sqrt{(1/4 + 3/4)} = 1$

$\tan \theta = (-\sqrt{3}/2)/(-1/2) = \sqrt{3} \Rightarrow$ reference angle $= \pi/3$

both components negative $\Rightarrow$ QIII, so $\theta = \pi + \pi/3 = 4\pi/3$

Find $(3\mathbf{u}) \cdot \mathbf{v}$ and $\mathbf{u} \cdot (\mathbf{u} + \mathbf{v})$ for the given vectors. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and state whether the angle between them is acute, right, or obtuse. (3 points)

5. $\mathbf{u} = -5\mathbf{i}, \mathbf{v} = 4\mathbf{i} - 3\mathbf{j}$

$$\mathbf{u} = \langle -5, 0 \rangle, \mathbf{v} = \langle 4, -3 \rangle$$

$$\mathbf{u} \cdot \mathbf{v} = (-5)(4) + (0)(-3) = -20$$

$$(3\mathbf{u}) \cdot \mathbf{v} = 3(-20) = \boxed{-60}$$

$$\mathbf{u} + \mathbf{v} = \langle -1, -3 \rangle \Rightarrow \mathbf{u} \cdot (\mathbf{u} + \mathbf{v}) = (-5)(-1) + (0)(-3) = \boxed{5}$$

$$\mathbf{u} \cdot \mathbf{v} = -20 \neq 0, \text{ so not orthogonal; negative} \Rightarrow \text{obtuse}$$

Find the angle between the given vectors. Round to the nearest tenth of a degree. (3 points)

6. $\mathbf{u} = 4\mathbf{i} - 3\sqrt{2}\mathbf{j}, \mathbf{v} = \sqrt{2}\mathbf{i} + 5\mathbf{j}$

$$\mathbf{u} \cdot \mathbf{v} = (4)(\sqrt{2}) + (-3\sqrt{2})(5) = 4\sqrt{2} - 15\sqrt{2} = -11\sqrt{2}$$

$$|\mathbf{u}| = \sqrt{(4)^2 + (-3\sqrt{2})^2} = \sqrt{16 + 18} = \sqrt{34}, |\mathbf{v}| = \sqrt{(\sqrt{2})^2 + (5)^2} = \sqrt{27}$$

$$\cos \theta = (-11\sqrt{2}) / (\sqrt{34}\sqrt{27}) \approx -0.5134$$

$$\theta = \cos^{-1}(-0.5134) \approx \boxed{120.9^\circ}$$

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force. Round to the nearest tenth. (3 points)

7. forces of 88 N and 130 N, forming an angle of 71°

$$\text{angle across from the diagonal: } 180^\circ - 71^\circ = 109^\circ$$

$$x = \sqrt{(88^2 + 130^2 - 2(88)(130)\cos 109^\circ)}$$

$$x \approx \boxed{179.1 \text{ N}}$$